Hitbhai Lunagariya

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Profile

A third-year B.Tech Student in Engineering Physics, SVNIT, with practical experience in various stages of AI/ML pipeline - from building data pipelines, models training (**YOLOv8, PyTorch, OpenCV**) to deploying on embedded systems (**ROS2, OpenVINO, ESP32**). Ranked **9th internationally** at IRC'26, which was the only team representing NIT, and developed a **RAG based** report generation pipeline using (Python, LangChain, Qdrant) as part of my internship in AI/ML at EnviroWealth.

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT) Aug 2024 – May 2028 (Expected)
B.Tech in Engineering Physics *CGPA: 9.34/10 (Rank 1)*
Minor in Artificial Intelligence *CGPA: 10/10*

Experience

EnviroWealth | AI/ML Intern | *Tesseract, LangChain, Streamlit, Qdrant* May 2026 – Jul 2026

- Contributed to a **RAG-based pipeline** for generating reports (OCR using **Tesseract**, chunking using Docling, vector database using **Qdrant**) for automated generation of BRSR sustainability reports
- Built a standalone chatbot prototype using **Streamlit** and the **Gemini API** to explore conversational report generation.

Projects

International Rover Challenge (IRC'26) | *Python, OpenCV, YOLOv8, ROS2* Oct 2025 – Feb 2026

- Developed a real-time computer vision pipeline for the IRC'26 Reconnaissance and Delivery Operation (RADO) mission, enabling fully autonomous cone detection, classification, and navigation on a rover.
- Expanded a Kaggle dataset from **4** to **6 cone classes** via **HSV-based synthetic generation**,trained a custom **YOLOv8** achieving reliable **mAP** across varied lighting conditions.
- Optimized YOLOv8 model via **OpenVINO** conversion for CPU-only deployment on Intel NUC, resolving real-time latency bottleneck and enabling reliable real-time cone tracking.
- Built an **ArUco marker detection** and pose estimation system for rover navigation, in the Autonomous Mission.
- Deployed the model inside a Robot Operating System (**ROS2**) to stream cone coordinates to the navigation stack.

Real-Time Gesture Controlled Robot | *Python, OpenCV, ESP32, CNN* Jul 2025 – Oct 2025

- Designed and built a robot controlled through real-time hand gesture recognition using a neural network model (YOLOv11x) capable of identifying 8 gesture classes with high accuracy.
- Implemented an OpenCV image processing pipeline; trained for **50 epochs** on a **2,000+ image** dataset with augmentation to handle varied hand positions and environments.
- Programmed a microcontroller (ESP32) via Arduino IDE for motor control — optimized end-to-end detection to movement latency to under **100ms** for real-time response.

Technical Skills

Core Areas: Computer Vision, Object Detection, Deep Learning, CNN, Machine Learning, Autonomous Systems
Frameworks & Libraries: OpenCV, YOLOv8, Langchain, LangGraph, PyTorch, TensorFlow, OpenVINO
Programming Languages: Python, C, C++
Robotics & Embedded: Robot Operating System (ROS2), ESP32, Arduino, LiDAR, SolidWorks

Leadership & Achievements

International Rover Challenge 2026 (IRC'26) — 9th Internationally
Branch Councilor – Engineering Physics, SVNIT (2024–2025)
Member – Drishti Technical Chapter (2025–Present)
Executive – Alumni Training & Placement Affairs Council SVNIT (2025–Present)